Brightlearn Data analytics

Research Assignment 1

1. Database
- an organized collection of data that is stored electronically so it can be easily accessed, Managed and updated

Databases are essential because
- they allow fast access to information
- Store large amounts of data efficiently
- Support decision macing

E.g - Bancing systems
- E-commerce websites

2. Data warehouse
- is a large system used to store historical data from different sources so it can be analyzed and use for reporting and decision macing

- Data base = for running the business
Data warehouse = for analyzing the business

a company uses a data base
- Mace better decisions
- Combine data from different sources
- Store historical data

3. Data Lake
- A system that stores large amounts of raw data in its original format

Data Lake = raw, flexible storage
Data warehouse = Clean, organized data for reporting

Businesses prefers using a Data Lake
- they have huge volumes of data
- They need flexibility
- Handling unstructured Data

4. Data Mart
- A smaller, focused subset of a data warehouse that is designed for specific department or business function

it relates to data warehouse
- a data warehouse - Everything is inside
Data Mart = a specific aisle

- Finance team typically uses it
- Sales team typically uses it

7. ~~Minii~~ Data Mining
- is the process of analyzing large amounts of data to discover patterns, trends and useful insights

~~Eg~~- it used in business
- mace better decisions
- understand customer behaviour

eg - Retail (customer buying patterns)
- Banking (fraud detection)

8. Database Management system
Software used to create, store, manage and retrieve Data in a data base

popular platforms
- My$ SqL - open source relational database system that used SQ
eg web application

Microsoft SqL Server
a powerful enterprise - level relational DBMS developed microsoft

Oracle database
a high performance, enterprise - grade DBMS known for security and scalability

9. Etract, Transform, Load
- Data integration process used to move data from different sources into a system where it can be analyzed, usually a data warehouse

- Extract
first step, where data is collected from different sources
e.g - online store system
- payment system
- customer database

- Transform
- it is all about cleaning and preparing the data
◦ Removing duplicates
- fixing errors

e.g - converting "Rando"
- fixing mixing customer names

Load
- final step where cleaned data is stored in a target system
- Data warehouse
- Data mart
- Cloud database

e.g - Clean sales data

ETL is important in Data analytics
- improves data quality
- combine data from multiple sources
- supports better decision making

10. in data analytics, data is stored in different formats depending on how it will be used

1. CSV
Simple data storage
Excel and spreadsheets
- Small data sets

2. JSON
flexible formats that stores data in key-value pairs often nested
- web applications
- Apis

Parquet
Column-based storage format designed for big data and analytics
- Large data sets
- Data warehouses

Avro
a row-based binary format that stores data long with its schema

11. ~~Artificia~~ Artificial intellegence
the computer science that focuses on Creating Systems that can perform tasks that normally require human intelligence

Narrow AI
is designed to perform one specific task only
e.g Google Maps

General AI
AI that can think, learn and understand like a human accross different tasks
e.g Study finance
- write code
- Drive a car

12 Machine Learning
a branch of artificial intellegence where computers are trained to learn patterns from data and make predictions

Traditional programming
- Human writes rules and instructions

Supervised Learning
Trained using labelled data
- out put
- ~~input~~ in put

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Unsupervised Learning
The model is given unlabelled data and must find patterns on its own

Reinforcement Learning
model learns by trial and error, receiving rewards or penalties

13. Deep Learning
Subfield of machine learning that used neutral networks with many layers to learn from large amount of data and make decisions

Artificial intillegence
The broad field of creating machines that can perform tasks that require human intillegence (like thinking, learning and problem solving)

Machine learning
Subset of AI that allows system to learn from data without being explicit programmed

Neutral Networks
- input layers - Receives data
- Hidden layers - process the data using weights and mathematical
Output layer - produces the final results

14. Natural Language Processing
A branch of artificial intilligence that helps computers, understand, interpret and respond to human language

Three real world applications
- Chatbots
- voice assistants
- Translation tools like google translate

15 Large Language Model
An advanced AI Model that understands and generates human language

Standard Machine learning
-handle language (text) specifically
- they use deep learning and very large data sets

16. Generative AI
a type of artificial intillegence that creates new content
- ChatGPT - generates text and answers
- Dall.E - Creates images from text
Midjourney - generates artistic images

17. Training Datasets
collection of data used to train an AI Model
- high-quality data
- poor quality data

Unsupervised
The model is
must find p

Reinforcement
model learns
rewards or p

13. Deep Le
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Machine
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Neutral
- input
- Hidde

Output

~~what could go wrong~~
~~a collection of data used to train an Ai model~~
- Biased data
- incorrect data
- Missing data

18. computer vision
A field of artificial intelligence that enables computers to see, analyze and understand images and videos

18. images are converted into pixel data
- the model compares patterns with learned data

19. Application programming interface
an Api is a set of rules that allows different software systems to communicate and share data

Simple real-world analogy
- You (the user) place an order
- The waiter (Api) takes your request to the kitchen
- The kitchen prepares the food
- The waiter brings the results back to you. The Api connects you to the service without you needing to know how it works inside

20 API Key is a unique Code used to identify and authenticate a user or application when accessing an API
- It is Purpose to Control access, track usage, and ensure only authorized users can use the Service.

* Storage and protection
- Store it in environment variables, not in your code
- Never share it publicly.

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* If exposed Publicly
- Unathorized users can use the API.
- Data could be stolen or misused.

21 A REST API is a type of API that follows the principles of Rest, allowing Communication between Systems over HTTP
- REST Stand for Representational State Transfer

* Main HTTP methods
- Delete - Remove a resource
- Put - update an existing resource.

22 JSON is a lightweight format used to store and exchange data

* Why it is Commonly used
- Easy for humans to read and write.
- Easy for machines to Parse.
- Work well with APIs.